Conformalized 1-Bit Matrix Completion

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***Abstract*—Matrix completion from binary observations is a fundamental problem in modern data science with wide applications. While many effective estimation algorithms exist in the literature, uncertainty quantification for this problem has proved to be challenging, and existing asymptotic methods are extremely sensitive to model misspecification and finite-sample limitations. In this work, we propose a distribution-free method for predictive inference in the 1-bit matrix completion problem. Our method adapts the framework of weighted conformal prediction, which provides confidence intervals with guaranteed distribution-free validity, to the problem of 1-bit matrix completion. Our resulting method, Conformalized 1-Bit Matrix Completion (C1B-MC), offers provable finite-sample marginal coverage regardless of the accuracy of the underlying parametric models or the exact low-rank assumptions. Empirical results on simulated and real data demonstrate that C1B-MC is robust to severe model misspecification and over-parameterization, matching the target coverage reliably while existing Fisher-information-based methods fail catastrophically.**

***Index Terms*—Matrix Completion, Conformal Prediction, Uncertainty Quantification, Missing Data, Trustworthy Recommender Systems.**

I. Introduction

Matrix completion is a cornerstone of modern data analysis with applications ranging from collaborative filtering [1] to causal panel data models [2]. In the context of recommender systems, uncertainty quantification (UQ) is critical for building *trustworthy* platforms: it enables models to abstain from unreliable predictions, thereby improving robustness against adversarial users and enhancing explainability. A particularly important variant is *1-bit matrix completion* [3], where only binary observations are available: a user likes or dislikes a product, or a student answers a question correctly or incorrectly. Formally, we observe a binary matrix $\boldsymbol{Y} \in \{-1, +1\}^{m\times n}$ whose entries are linked to an underlying real-valued latent matrix $\boldsymbol{M}^\star$ through a known probabilistic link function $\mathbb{P}(Y_{ij} = 1 \mid M^\star_{ij}) = \sigma(M^\star_{ij})$. Given an observation mask $\boldsymbol{\Omega} \in \{0, 1\}^{m\times n}$, the goal is to recover $\boldsymbol{M}^\star$ from $\{Y_{ij} : \Omega_{ij} = 1\}$.

While point estimation for this problem has been explored extensively [3]–[5], uncertainty quantification remains challenging. A recent framework by Chen et al. [6] provides asymptotic confidence intervals based on plug-in Fisher information. However, the validity of such asymptotic methods relies heavily on three critical assumptions: correct model specification, large enough matrix dimensions, and uniform observation patterns. In practice, these assumptions are often violated (e.g., heavy-tailed noise, non-uniform ratings), causing asymptotic coverage to degrade drastically—from the nominal 90% to below 30% in the worst case (Section V). This motivates our core question: *can we construct valid, finite-sample confidence intervals for latent entries from binary observations that are robust to model misspecification?*

A. Conformal prediction for distribution-free uncertainty quantification

The conformal prediction framework [7]–[9], particularly its weighted variant for covariate shift [10], offers a compelling distribution-free alternative. Recently, Gui et al. [11] introduced *Conformalized Matrix Completion* (CMC), successfully adapting weighted conformal prediction to standard continuous matrix completion. Similar robust conformal adaptations have also recently been explored in related matrix problems, such as robust principal component analysis [12].

Extending this framework to the 1-bit setting, however, raises several non-trivial challenges. First, the link function σ introduces a fundamental non-linearity that changes the structure of the likelihood. More critically, unlike the continuous case where the initial estimate $\widehat{\boldsymbol{M}}$ can be evaluated directly on calibration entries, in the 1-bit setting the latent matrix $\boldsymbol{M}^\star$ is never directly observed. This means the ideal nonconformity score $V_{ij} = |\widehat{\boldsymbol{M}}_{ij} - M^\star_{ij}|/\boldsymbol{S}_{ij}$ is inaccessible. We must therefore account for the estimation error propagation—bounding the additional error from using the fitted $\widehat{\boldsymbol{M}}$ (often derived from constrained non-convex optimization) in place of $\boldsymbol{M}^\star$.

B. Main contributions

In this paper, we adapt conformal prediction to 1-bit matrix completion, making the following contributions:

1) We propose **Conformalized 1-Bit Matrix Completion (C1B-MC)** (Algorithm 1), a framework that produces distribution-free, finite-sample valid confidence intervals for latent matrix entries, regardless of the correctness of the assumed parametric link function.
2) We prove a **finite-sample marginal coverage guarantee** (Theorem 4.3). We explicitly decompose the coverage gap into a covariate-shift term δ_{exch} (from non-uniform observations) and an estimation error term δ_{est} (from the discrepancy between $\widehat{\boldsymbol{M}}$-based and $\boldsymbol{M}^\star$-based scores), bounding the latter via existing matrix completion theory.
3) To ensure **real-data applicability** where $\boldsymbol{M}^\star$ is entirely unknown, we develop a practical variant of C1B-MC that reformulates the calibration process using hinge scores based purely on the observable binary labels $\boldsymbol{Y}$.
4) Through **extensive experiments** (including the MovieLens 100K dataset), we demonstrate that C1B-MC

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reliably achieves the target coverage ($\approx 90\%$), explicitly correcting the catastrophic failures of asymptotic methods under severe misspecification and overparameterization.

Our open-source code implementation is publicly available at https://github.com/xingyueqing1-source/C1B-MC. Furthermore, we provide an Appendix detailing convex vs. nonconvex estimation approaches and a jackknife+ full conformal variant.

II. Problem Setup

A. The 1-Bit Matrix Completion Model

Let $\boldsymbol{M}^\star \in \mathbb{R}^{m\times n}$ be an unknown latent matrix. We assume $\boldsymbol{M}^\star$ is approximately low-rank, with effective rank $r \ll \min(m,n)$. In contrast to standard continuous matrix completion where entries are observed directly with additive noise, in the 1-bit setting we only observe binary labels $Y_{ij} \in \{-1,+1\}$ drawn from a probabilistic model $\mathbb{P}(Y_{ij}=1 \mid M^\star_{ij}) = \sigma(M^\star_{ij})$. Here, $\sigma:\mathbb{R}\to[0,1]$ is a known link function. We primarily focus on the logistic link $\sigma(x) = 1/(1+e^{-x})$, though our framework seamlessly accommodates any differentiable, monotone alternative (e.g., the probit link). Following [6] and [5], we impose a boundedness constraint on the latent matrix: $\|\boldsymbol{M}^\star\|_\infty \le \alpha$, where $\alpha > 0$ is a known constant. This ensures that the Bernoulli probabilities are bounded away from 0 and 1, a condition necessary for identifiability and for the validity of Fisher-information-based inference.

B. Observation Mechanism and Assumptions

The entries of the matrix are observed according to a random mask $\boldsymbol{\Omega} \in \{0,1\}^{m\times n}$. Let $\boldsymbol{P}_{ij} = \mathbb{P}(\Omega_{ij}=1)$ denote the observation probability for entry (i,j). The observed data consist of $\mathcal{D} = \{(i,j,Y_{ij}) : \Omega_{ij}=1\}$. We make the following structural assumptions, which are standard in the matrix completion literature:

Assumption 2.1 (Row-Column Exchangeability): The rows (users) and columns (items) are randomly ordered. Conditioned on the latent parameters generating $\boldsymbol{M}^\star$, the pairs $\{(i,j) : \Omega_{ij}=1\}$ are exchangeable across entries, subject to the observation probability matrix $\boldsymbol{P}$.

Assumption 2.2 (Observation Model): For each (i,j), $\Omega_{ij} \sim \text{Bernoulli}(\boldsymbol{P}_{ij})$ independently, and $Y_{ij} \sim \text{Bernoulli}(\sigma(M^\star_{ij}))$ independently, with the missingness mechanism satisfying $\Omega_{ij} \perp Y_{ij} \mid M^\star_{ij}$.

Inference Goal. Our primary goal is to provide uncertainty quantification for the unobserved latent entries. Specifically, given a target miscoverage level $\alpha_{\text{cp}} \in (0,1)$ (distinct from the boundedness constant α), we aim to construct a marginal confidence interval C_{ij} for each test entry such that $\mathbb{P}(M^\star_{ij} \in C_{ij}) \ge 1-\alpha_{\text{cp}}$.

C. The Core Challenge: Latent Targets and Dual Sources of Error

A critical distinction between our setting and standard conformalized matrix completion [11] is that the inference target $\boldsymbol{M}^\star$ is *never directly observed*. In standard conformal inference, the target is the observable label Y_{ij}. Here, to achieve granular entry-wise inference, we define our ideal nonconformity score as the normalized residual $V^{\text{ideal}}_{ij} = |\widehat{\boldsymbol{M}}_{ij} - M^\star_{ij}|/\boldsymbol{S}_{ij}$.

The local standard error $\boldsymbol{S}_{ij}$ serves as a plug-in estimate of the asymptotic variance of $\widehat{\boldsymbol{M}}_{ij}$ derived from Fisher information. In the context of Chen et al. [6], who consider inference for general linear functionals $g(\boldsymbol{M}^\star)$, our score implicitly selects the identity functional $g_{ij}(\boldsymbol{M}^\star) = M^\star_{ij}$, representing the most granular form of inference.

Because the ideal score relies on the unobservable $M^\star_{ij}$ during the calibration stage, we must substitute it with a feasible proxy (based on $\boldsymbol{Y}$ or $\widehat{\boldsymbol{M}}$). This substitution introduces a "second source of error" beyond the covariate-shift caused by nonuniform sampling. Consequently, as we will formally establish in Section IV, our overall coverage guarantee must explicitly account for both gaps: $\mathbb{P}(M^\star_{ij} \in C_{ij}) \ge 1-\alpha_{\text{cp}} - \delta_{\text{exch}} - \delta_{\text{est}}$, where δ_{exch} accounts for the covariate-shift discrepancy (from weighted exchangeability), and δ_{est} accounts for the estimation error gap (from using proxy scores instead of ideal scores).

III. Conformalized 1-Bit Matrix Completion

We now present the Conformalized 1-Bit Matrix Completion (C1B-MC) framework. Following the standard split conformal methodology, we randomly partition the observed entries (indexed by $\boldsymbol{\Omega}$) into three disjoint sets: a training set Ω_{train}, a calibration set Ω_{cal}, and a test set Ω_{test} (e.g., in a $60\%/20\%/20\%$ ratio). Our procedure operates in three stages: point estimation on Ω_{train}, local uncertainty quantification, and weighted conformal calibration using Ω_{cal}. The complete procedure is summarized in Algorithm 1.

A. Stage 1: Latent Matrix Estimation

Given the training mask Ω_{train} and the observed binary entries $\boldsymbol{Y}$, we maximize the logistic log-likelihood under a log-barrier penalty that enforces the boundedness constraint $\|\boldsymbol{M}^\star\|_\infty \le \alpha$. Parameterizing the matrix as $\boldsymbol{M} = \boldsymbol{U}\boldsymbol{V}^\top$ with $\boldsymbol{U} \in \mathbb{R}^{m\times k}$ and $\boldsymbol{V} \in \mathbb{R}^{n\times k}$, we solve the following optimization problem:

$$\min_{\boldsymbol{U},\boldsymbol{V}} \sum_{(i,j)\in\Omega_{\text{train}}} \log\left(1+\exp\left(-Y_{ij}(\boldsymbol{U}\boldsymbol{V}^\top)_{ij}\right)\right) - \lambda \sum_{i,j} \log\left(1-\frac{(\boldsymbol{U}\boldsymbol{V}^\top)^2_{ij}}{\alpha^2}\right), \tag{1}$$

where $\lambda > 0$ controls the barrier strength. The log-barrier term acts as a soft constraint, penalizing entries as they approach $\pm\alpha$. We optimize (1) via gradient descent with backtracking line search, selecting λ through cross-validation.

Remark 3.1 (Non-convexity). The bilinear factorization $\boldsymbol{M} = \boldsymbol{U}\boldsymbol{V}^\top$ renders (1) non-convex. This choice is deliberate: the non-convex formulation is computationally more efficient and often yields empirically superior point estimates compared to convex relaxations (see Appendix). Crucially, the subsequent conformal calibration in Stage 3 rigorously corrects for any downstream effects of potential sub-optimality in this point estimate.

B. Stage 2: Local Uncertainty Quantification

To adaptively scale our conformal intervals, we estimate the entry-wise asymptotic standard error based on the Fisher information analysis of Chen et al. [6]. For the logistic model, the conditional variance is given by $\mathrm{Var}(Y_{ij} \mid M^\star_{ij}) = \sigma(M^\star_{ij})(1-\sigma(M^\star_{ij}))$. We define the estimated local standard error at entry (i,j) as:

$$\boldsymbol{S}_{ij} = \sqrt{\frac{1}{\sum_\ell \hat{\sigma}^2_{i\ell}\mathbf{1}_{\{(i,\ell)\in\Omega_{\text{train}}\}}+\epsilon} + \frac{1}{\sum_k \hat{\sigma}^2_{kj}\mathbf{1}_{\{(k,j)\in\Omega_{\text{train}}\}}+\epsilon}} \quad (2)$$

where $\hat{\sigma}^2_{ij} = \sigma(\widehat{\boldsymbol{M}}_{ij})(1-\sigma(\widehat{\boldsymbol{M}}_{ij}))$ is clamped to $[10^{-4}, 0.25]$ to prevent vanishing denominators, and $\epsilon = 10^{-6}$ ensures numerical stability.

C. Stage 3: Weighted Conformal Calibration

Because the observation probabilities $\boldsymbol{P}_{ij}$ may vary across entries, the calibration and test entries are not exchangeable. To correct for this covariate shift, we employ the odds-ratio weighting scheme introduced in [10]. For the calibration entries $(i',j') \in \Omega_{\text{cal}}$ and a given test entry $(i,j) \in \Omega_{\text{test}}$, we define the weights as:

$$w_{i'j'} = \frac{1-\boldsymbol{P}_{i'j'}}{\boldsymbol{P}_{i'j'}}, \qquad w_{\text{test}} = \frac{1-\boldsymbol{P}_{ij}}{\boldsymbol{P}_{ij}}. \quad (3)$$

Oracle Calibration (Simulation). In synthetic settings where the true $\boldsymbol{M}^\star$ is known, we define the normalized residual score $V_{ij} = |\widehat{\boldsymbol{M}}_{ij} - M^\star_{ij}|/\boldsymbol{S}_{ij}$ for $(i,j) \in \Omega_{\text{cal}}$. The conformal quantile Q_{ij} for a test entry (i,j) is the smallest value q such that the normalized cumulative weight reaches the $1-\alpha_{\text{cp}}$ level:

$$Q_{ij} = \inf\left\{ q : \frac{\sum_{(i',j')\in\Omega_{\text{cal}}} w_{i'j'}\mathbf{1}_{\{V_{i'j'}\le q\}} + w_{\text{test}}\cdot\mathbf{1}_{\{\infty\le q\}}}{\sum_{(i',j')\in\Omega_{\text{cal}}} w_{i'j'} + w_{\text{test}}} \ge 1-\alpha_{\text{cp}} \right\}. \quad (4)$$

The $(1-\alpha_{\text{cp}})$-level confidence interval for $M^\star_{ij}$ is then constructed as $C_{ij} = [\widehat{\boldsymbol{M}}_{ij} - Q_{ij}\boldsymbol{S}_{ij}, \widehat{\boldsymbol{M}}_{ij} + Q_{ij}\boldsymbol{S}_{ij}]$. We deliberately apply no post-processing truncation to these intervals, as the conformal guarantee holds for the untruncated sets. Conformal p-values can also be inverted from these intervals for hypothesis testing (e.g., $H_0 : M^\star_{ij} = 0$).

D. Real-Data Variant: Calibration without True Latent Matrix

In real-world applications, $\boldsymbol{M}^\star$ is completely unobservable, rendering the residual score V_{ij} inaccessible. To provide actionable uncertainty quantification for the binary labels, we replace the latent residual score with a *hinge score* based on the observable labels $\boldsymbol{Y}$: $V^{\text{real}}_{ij} = -Y_{ij}\cdot\widehat{\boldsymbol{M}}_{ij}$. This score measures the binary classification margin: large positive values indicate agreement (same sign), while large negative values indicate disagreement. The weighted quantile Q_{ij} is computed identically via (4) using $V^{\text{real}}_{i'j'}$, and the prediction set for the binary test label Y_{ij} is $\widehat{C}^{\text{pred}}_{ij} = \{y \in \{-1,+1\} : -y\cdot\widehat{\boldsymbol{M}}_{ij} \le Q_{ij}\}$. If $|\widehat{\boldsymbol{M}}_{ij}| > Q_{ij}$, the set collapses to a definitive singleton $\{\mathrm{sign}(\widehat{\boldsymbol{M}}_{ij})\}$; otherwise, it outputs the full set $\{-1,+1\}$, gracefully abstaining from prediction when uncertainty is high.

Computational Cost and Scalability. A key advantage of C1B-MC is its computational efficiency, which is vital for large-scale recommender systems that require rapid responses. C1B-MC requires only a single model training run. The conformal calibration step introduces minimal overhead, scaling as $\mathcal{O}(|\Omega_{\text{cal}}|\log|\Omega_{\text{cal}}|)$ to compute the weighted quantiles. This makes inference highly scalable compared to ensemble-based conformal methods (e.g., the jackknife+, see Appendix B).

Algorithm 1 Conformalized 1-Bit Matrix Completion (C1B-MC)

Require: Binary observations $\boldsymbol{Y}$, mask $\boldsymbol{\Omega}$, probability matrix $\boldsymbol{P}$, rank k, bound α, target level α_{cp}.
Ensure: Confidence intervals $\{C_{ij}\}$ (Simulation) or Prediction sets $\{\widehat{C}^{\text{pred}}_{ij}\}$ (Real Data).

1: **Data Split:** Partition observed entries into Ω_{train}, Ω_{cal}, and Ω_{test}.
2: **Point Estimation:** Fit $\widehat{\boldsymbol{M}} = \boldsymbol{U}\boldsymbol{V}^\top$ on Ω_{train} via gradient descent on (1).
3: **Local Uncertainty:** Compute local standard errors $\boldsymbol{S}_{ij}$ using (2) for all entries.
4: **for** each calibration entry $(i,j) \in \Omega_{\text{cal}}$ **do**
5: $\quad V_{ij} \leftarrow |\widehat{\boldsymbol{M}}_{ij} - M^\star_{ij}|/\boldsymbol{S}_{ij}$ {*Simulation (Oracle)*}
6: $\quad V_{ij} \leftarrow -Y_{ij}\cdot\widehat{\boldsymbol{M}}_{ij}$ {*Real Data Variant*}
7: **end for**
8: **for** each test entry $(i,j) \in \Omega_{\text{test}}$ **do**
9: $\quad$ Calculate odds-ratio weights $w_{i'j'}$ for all Ω_{cal} and w_{test} for (i,j).
10: $\quad$ Compute the weighted $(1-\alpha_{\text{cp}})$-quantile Q_{ij} using (4).
11: $\quad C_{ij} \leftarrow [\widehat{\boldsymbol{M}}_{ij} - Q_{ij}\boldsymbol{S}_{ij}, \widehat{\boldsymbol{M}}_{ij} + Q_{ij}\boldsymbol{S}_{ij}]$ {*Simulation interval*}
12: $\quad \widehat{C}^{\text{pred}}_{ij} \leftarrow \{y \in \{-1,+1\} : -y\cdot\widehat{\boldsymbol{M}}_{ij} \le Q_{ij}\}$ {*Real Data Variant*}
13: **end for**

IV. Theoretical Guarantees

The fundamental challenge in applying conformal prediction to matrix completion is that the calibration and test entries are not identically distributed when the observation probability is non-uniform. Furthermore, unlike standard regression, our ideal score relies on an unobserved latent target $\boldsymbol{M}^\star$. In this section, we explicitly bound both sources of error.

A. Weighted Exchangeability and the First Error

Tibshirani et al. [10] established that weighted conformal prediction with correctly specified likelihood-ratio weights $w(x) = \mathrm{d}P_{\text{test}}(x)/\mathrm{d}P_{\text{cal}}(x)$ provides valid coverage under covariate shift. In our setting, the likelihood ratio for an entry to be unobserved (test) versus observed (calibration) is exactly the odds of non-observation: $w(\boldsymbol{P}_{ij}) = (1-\boldsymbol{P}_{ij})/\boldsymbol{P}_{ij}$.

Lemma 4.1 (Weighted Exchangeability, adapting [10]): Under Assumptions 2.1 and 2.2, and given that the observation

probability matrix $\boldsymbol{P}$ is known (or consistently estimated), the weighted empirical distribution of the calibration scores,

$$\widehat{F}_{\text{cal}}(v) = \frac{\sum_{(i,j)\in\Omega_{\text{cal}}} w_{ij}\mathbf{1}_{\{V_{ij}\le v\}}}{\sum_{(i,j)\in\Omega_{\text{cal}}} w_{ij}}, \tag{5}$$

converges to the true distribution of the test scores $F_{\text{test}}(v)$ as $|\Omega_{\text{cal}}| \to \infty$. For finite $|\Omega_{\text{cal}}|$, the total variation distance is bounded by the covariate-shift error:

$$\delta_{\text{exch}} \le d_{\text{TV}}\left(\widehat{F}_{\text{cal}}, F_{\text{test}}\right) = \mathcal{O}\left(|\Omega_{\text{cal}}|^{-1/2}\right) + \sup_{i,j,i',j'} |\boldsymbol{P}_{ij} - \boldsymbol{P}_{i'j'}|. \tag{6}$$

The first term accounts for the standard Monte Carlo error from a finite calibration size, while the second term captures the systematic discrepancy when observation probabilities vary across the matrix. In the special case of uniform observation ($\boldsymbol{P}_{ij} \equiv p$), $\delta_{\text{exch}} = 0$, and standard (unweighted) split conformal prediction achieves exact coverage.

B. The Estimation Error (Second Error)

In simulation settings, we can compute the ideal score V_{ij}^{ideal} using the true $M_{ij}^{\star}$. In real applications, we substitute it with an estimate $\widehat{\boldsymbol{M}}_{ij}$ (or the observed binary label Y_{ij}). This substitution introduces a distributional shift. Let F_{ideal} denote the distribution of the ideal scores $V_{ij}^{\text{ideal}} = |\widehat{\boldsymbol{M}}_{ij} - M_{ij}^{\star}|/\boldsymbol{S}_{ij}$, and let $\widehat{F}_{\text{est}}$ be the distribution of the feasible proxy scores.

Lemma 4.2 (Estimation Error Bound): Assume the mapping from the latent value to the nonconformity score is L-Lipschitz. Let $\|\widehat{\boldsymbol{M}} - \boldsymbol{M}^{\star}\|_F$ be the Frobenius error of the point estimator. Then the Wasserstein-1 distance between the ideal and estimated score distributions satisfies:

$$\delta_{\text{est}} \le L \cdot \frac{\|\widehat{\boldsymbol{M}} - \boldsymbol{M}^{\star}\|_F}{\sqrt{|\Omega_{\text{cal}}|}\cdot \min_{i,j} \boldsymbol{S}_{ij}}. \tag{7}$$

Furthermore, under the exact low-rank model conditions of [5],

$$\|\widehat{\boldsymbol{M}} - \boldsymbol{M}^{\star}\|_F \le C \cdot \sqrt{\frac{r\cdot\max(m,n)\cdot\log(\max(m,n))}{|\Omega_{\text{train}}|}}, \tag{8}$$

where C depends on the boundedness parameter α and the link function σ.

Proof: Let V_{ij}^{ideal} be the ideal nonconformity score evaluated using the true latent entry $M_{ij}^{\star}$, and $\widehat{V}_{ij}$ be the feasible proxy score evaluated using the estimate $\widehat{\boldsymbol{M}}_{ij}$. By the L-Lipschitz assumption of the score function (scaled by the local standard error $\boldsymbol{S}_{ij}^{-1}$), we have for any entry (i,j):

$$|V_{ij}^{\text{ideal}} - \widehat{V}_{ij}| \le L \cdot \frac{|\widehat{\boldsymbol{M}}_{ij} - M_{ij}^{\star}|}{\boldsymbol{S}_{ij}}. \tag{9}$$

The Wasserstein-1 distance W_1 between the empirical distribution of the ideal scores ($\widehat{F}_{\text{ideal}}$) and the estimated scores ($\widehat{F}_{\text{est}}$) is upper-bounded by the average entry-wise absolute difference over the calibration set:

$$\begin{aligned} W_1(\widehat{F}_{\text{ideal}}, \widehat{F}_{\text{est}}) &\le \frac{1}{|\Omega_{\text{cal}}|}\sum_{(i,j)\in\Omega_{\text{cal}}} |V_{ij}^{\text{ideal}} - \widehat{V}_{ij}| \\ &\le \frac{L}{|\Omega_{\text{cal}}|\cdot\min_{i,j}\boldsymbol{S}_{ij}} \sum_{(i,j)\in\Omega_{\text{cal}}} |\widehat{\boldsymbol{M}}_{ij} - M_{ij}^{\star}|. \end{aligned} \tag{10}$$

Applying the Cauchy-Schwarz inequality to the sum yields:

$$\sum_{(i,j)\in\Omega_{\text{cal}}} |\widehat{\boldsymbol{M}}_{ij} - M_{ij}^{\star}| \le \sqrt{|\Omega_{\text{cal}}|}\cdot\sqrt{\sum_{(i,j)\in\Omega_{\text{cal}}} (\widehat{\boldsymbol{M}}_{ij} - M_{ij}^{\star})^2}. \tag{11}$$

Since the sum of squared errors over the calibration set Ω_{cal} is strictly bounded by the sum over all observed matrix entries, we inherently have $\sqrt{\sum_{\Omega_{\text{cal}}}(\widehat{\boldsymbol{M}}_{ij} - M_{ij}^{\star})^2} \le \|\widehat{\boldsymbol{M}} - \boldsymbol{M}^{\star}\|_F$. Substituting this back into the W_1 bound yields the first inequality:

$$\delta_{\text{est}} = W_1(\widehat{F}_{\text{ideal}}, \widehat{F}_{\text{est}}) \le L \cdot \frac{\|\widehat{\boldsymbol{M}} - \boldsymbol{M}^{\star}\|_F}{\sqrt{|\Omega_{\text{cal}}|}\cdot\min_{i,j}\boldsymbol{S}_{ij}}. \tag{12}$$

The second inequality bounding the Frobenius norm $\|\widehat{\boldsymbol{M}} - \boldsymbol{M}^{\star}\|_F$ follows directly from Bhaskar and Javanmard [5, Theorem 1] under their exact low-rank recovery constraints. ∎

C. Main Coverage Guarantee

Combining the bounds on the two sources of error, we establish the finite-sample validity of our proposed C1B-MC framework.

Theorem 4.3 (Finite-Sample Marginal Coverage): Under Assumptions 2.1 and 2.2, and with the weights $w(\boldsymbol{P}_{ij}) = (1-\boldsymbol{P}_{ij})/\boldsymbol{P}_{ij}$ correctly specified, the prediction intervals produced by Algorithm 1 satisfy, for any target level $\alpha_{\text{cp}} \in (0,1)$:

$$\mathbb{P}\left(M_{ij}^{\star} \in C_{ij}\right) \ge 1 - \alpha_{\text{cp}} - \delta_{\text{exch}} - \delta_{\text{est}}. \tag{13}$$

For the uniform observation case ($\boldsymbol{P}_{ij} \equiv p$) with exact scores ($\delta_{\text{est}} = 0$), the bound tightens to:

$$\mathbb{P}\left(M_{ij}^{\star} \in C_{ij}\right) \in \left[1 - \alpha_{\text{cp}},\ 1 - \alpha_{\text{cp}} + \frac{2}{|\Omega_{\text{cal}}| + 1}\right], \tag{14}$$

where the upper bound follows from standard split conformal guarantees [9].

Proof: Let E_{ideal} be the event that the true target $M_{ij}^{\star}$ is covered if we had access to the ideal latent scores V_{ij}^{ideal}. According to the weighted conformal prediction guarantees by Tibshirani et al. [10, Theorem 1], the coverage probability of this ideal event is lower bounded by the target level minus the total variation distance caused by covariate shift:

$$\mathbb{P}(E_{\text{ideal}}) \ge 1 - \alpha_{\text{cp}} - d_{\text{TV}}\left(\widehat{F}_{\text{cal}}, F_{\text{test}}\right) = 1 - \alpha_{\text{cp}} - \delta_{\text{exch}}. \tag{15}$$

In practice, we substitute the ideal scores with estimated scores, which induces a distributional shift bounded by the Wasserstein-1 distance δ_{est} (as derived in Lemma 4.2). Using the triangle inequality for probability measures, the difference

in coverage probability between the ideal conformal interval and our empirical conformal interval C_{ij} is bounded by the maximum probability mass shifted by this score estimation error:

$$\mathbb{P}\left(M^\star_{ij} \in C_{ij}\right) \geq \mathbb{P}(E_{\text{ideal}}) - \delta_{\text{est}}. \tag{16}$$

Combining these two bounds directly yields $\mathbb{P}\left(M^\star_{ij} \in C_{ij}\right) \geq 1 - \alpha_{\text{cp}} - \delta_{\text{exch}} - \delta_{\text{est}}$.

For the special case of uniform observations ($\boldsymbol{P}_{ij} \equiv p$), the calibration and test distributions are exactly exchangeable, meaning the weights are uniform and $\delta_{\text{exch}} = 0$. If we also have access to exact scores ($\delta_{\text{est}} = 0$), the standard split conformal argument [7], [9] applies directly, providing the exact finite-sample coverage interval $[1 - \alpha_{\text{cp}}, 1 - \alpha_{\text{cp}} + 2/(|\Omega_{\text{cal}}| + 1)]$. ∎

Remark on the upper bound. The conformal coverage explicitly features both a lower and an upper bound. The upper bound $1 - \alpha_{\text{cp}} + 2/(|\Omega_{\text{cal}}| + 1)$ mathematically proves that the coverage cannot be overly conservative—it remains within $\mathcal{O}(1/|\Omega_{\text{cal}}|)$ of the target. This is a profound advantage over naive Bonferroni-style union bounds, which often produce coverage arbitrarily close to 100% at the cost of infinite interval widths. For instance, in our experimental setup ($N = 500, p = 0.2$), $|\Omega_{\text{cal}}| \approx 10,000$, resulting in an upper bound of 0.9002, which precisely aligns with our empirical observations (e.g., 0.900 in Table 1).

V. Simulation Studies

We conduct numerical experiments to verify the finite-sample coverage guarantees of Conformalized 1-Bit Matrix Completion (C1B-MC) by dividing our experiments into two main parts: (1) **synthetic simulations**, designed to rigorously test the method under various observation mechanisms (homogeneous and heterogeneous) and stress-test its robustness against severe model misspecification; and (2) a **real-world application** on the MovieLens 100K dataset.

A. Synthetic Dataset

Data generation. Latent matrices $\boldsymbol{M}^\star$ are generated under two paradigms: (1) **General low-rank:** $\boldsymbol{M}^\star = \boldsymbol{U}\boldsymbol{V}^\top$, $\boldsymbol{U} \in \mathbb{R}^{m\times r}$, $\boldsymbol{V} \in \mathbb{R}^{n\times r}$ with i.i.d. Uniform$(-0.5, 0.5)$ entries, scaled so $\|\boldsymbol{M}^\star\|_\infty = \alpha$; (2) **Rasch:** $\boldsymbol{M}^\star_{ij} = \theta_i - \beta_j$, $\theta_i, \beta_j \sim$ Uniform$(-1, 1)$, centered. Default parameters are $m = n = N = 500$, true rank $r = 5$, observation rate $p = 0.2$, calibration ratio $\rho = 0.2$, $\alpha = 1.0$, target level $\alpha_{\text{cp}} = 0.10$. Each configuration is repeated over 3 independent trials.

Methods compared: We compare **C1B-MC (Ours)** (using normalized residual scores and odds-ratio weights) against **Asym [6]** (asymptotic z-intervals $\widehat{\boldsymbol{M}}_{ij} \pm z_{1-\alpha_{\text{cp}}/2} \cdot \boldsymbol{S}_{ij}$ based on plug-in Fisher information). The target empirical coverage is 0.90.

1) Homogeneous Missingness: In our first set of evaluations, entries are observed uniformly at random. We design three specific settings to systematically stress-test the assumptions of asymptotic inference:

- **Setting 1 (Well-specified):** Rasch model with logistic noise. All modeling assumptions are satisfied (the "home field" of asymptotic inference).
- **Setting 2 (Misspecified):** General low-rank model with heavy-tailed noise (t_2 perturbation before sigmoid). The noise distribution is mis-specified.
- **Setting 3 (Over-parameterization stress test):** General model with relaxed boundedness ($\alpha = 5.0$), weak regularization ($\lambda = 0.001$), and heavily over-parameterized rank ($r_{\text{fit}} = 12 \gg r_{\text{true}} = 5$).

TABLE I
SETTING 1: RASCH + LOGISTIC ($N = 500$, $p = 0.2$, $r_{\text{TRUE}} = 5$). COVERAGE AND INTERVAL LENGTH AS A FUNCTION OF HYPOTHESIZED RANK r_{FIT}.

	Coverage			**Avg. Length**		
r_{fit}	C1B-MC	Asym	Gap	C1B-MC	Asym	Ratio
2	0.900	0.773	+0.127	0.614	0.648	0.95
3	0.901	0.773	+0.128	0.617	0.649	0.95
5	0.900	0.775	+0.125	0.617	0.671	0.92
8	0.899	0.770	+0.129	0.621	0.666	0.93
12	0.898	0.773	+0.125	0.612	0.642	0.95

Setting 1 Results. Table I reports results for the well-specified model. C1B-MC consistently achieves the nominal 90% coverage across all hypothesized ranks. The asymptotic intervals under-cover at $\approx 77\%$—a gap of ≈ 13 percentage points—despite the model being correctly specified. This persistent under-coverage at finite $N = 500$ suggests that the central limit approximation requires substantially larger sample sizes to be accurate.

TABLE II
SETTING 2: GENERAL + HEAVY-TAIL ($N = 500$, $p = 0.2$, $r_{\text{TRUE}} = 5$). COVERAGE AND INTERVAL LENGTH BY r_{FIT}.

	Coverage			**Avg. Length**		
r_{fit}	C1B-MC	Asym	Gap	C1B-MC	Asym	Ratio
2	0.900	0.983	−0.083	0.716	1.047	0.68
3	0.900	0.983	−0.083	0.716	1.047	0.68
5	0.900	0.984	−0.084	0.715	1.047	0.68
8	0.899	0.983	−0.084	0.720	1.047	0.69
12	0.900	0.983	−0.083	0.716	1.047	0.68

Setting 2 Results. Table II shows results under misspecification. Here the heavy-tailed noise inflates the true variability, causing the asymptotic intervals to *over-cover* at $\approx 98\%$ but with intervals approximately 46% wider than C1B-MC's. The conformal method empirically self-calibrates to the true data-generating distribution, producing intervals that are both valid (exactly at target) and substantially tighter.

Setting 3 Results. Table III presents the stress test. The relaxed boundedness ($\alpha = 5.0$) and weak penalty drive the estimated Bernoulli probabilities close to 0 or 1. The plug-in Fisher information becomes artificially small, leading Asym coverage to crash to 28%. C1B-MC detects and corrects for this overconfidence, remaining immune to failures of the point estimator's uncertainty quantification.

Visual Summary and Score Analysis. Fig. 1 visualizes the performance across all hypothesized ranks, clearly showing

TABLE III
SETTING 3: STRESS TEST ($N = 500$, $p = 0.3$, $\alpha = 5.0$, $r_{\text{TRUE}} = 5$, $r_{\text{FIT}} = 12$). THE ASYMPTOTIC METHOD COLLAPSES.

	Coverage			Avg. Length		
r_{fit}	C1B-MC	Asym	Gap	C1B-MC	Asym	Ratio
2	0.902	0.364	+0.538	3.374	0.927	3.64
5	0.899	0.319	+0.580	3.916	0.956	4.10
8	0.899	0.300	+0.599	4.270	0.987	4.33
12	0.899	0.281	+0.618	4.417	1.005	4.39

that C1B-MC maintains a flat ≈ 0.90 coverage while Asym systematically deviates. To understand *why* asymptotic intervals fail under misspecification, Fig. 2 examines the empirical distribution of standardized residuals $(\widehat{M}_{ij} - M^{\star}_{ij})/\boldsymbol{S}_{ij}$. While the well-specified setting yields near-Gaussian residuals, the misspecified setting exhibits heavy tails and skewness. C1B-MC requires no distributional assumption and calibrates empirically from this observed tail behavior.

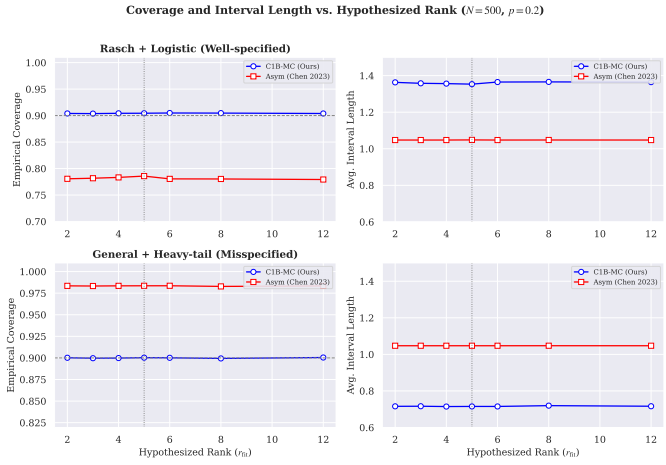


Fig. 1. Coverage and length vs. hypothesized rank ($N = 500$, $p = 0.2$). C1B-MC coverage is perfectly calibrated across all ranks.

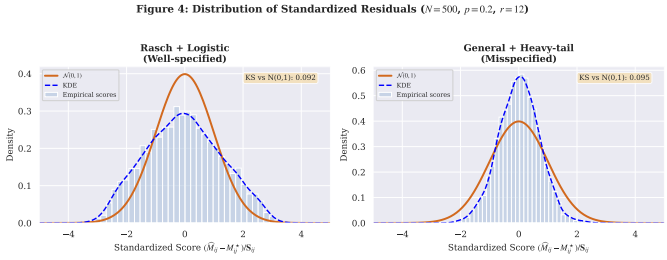


Fig. 2. Standardized residual distributions ($r_{\text{fit}} = 12$). **Left:** Well-specified setting, residuals closely follow $\mathcal{N}(0, 1)$. **Right:** Misspecified setting, residuals exhibit heavy tails. The conformal quantile adapts automatically; the asymptotic z-quantile does not.

2) Heterogeneous Missingness: To explicitly demonstrate the necessity and efficacy of the odds-ratio weighting scheme (Stage 3 of Algorithm 1), we conduct an experiment under heterogeneous missingness. In real-world scenarios, observations are rarely uniform; popular items and active users dominate the dataset, creating a severe covariate shift between the observed calibration set and the unobserved test set.

Setup. We simulate a heavily skewed observation mechanism on an $N = 500$ matrix with true rank $r = 5$ and heavy-tailed noise. The observation probabilities $\boldsymbol{P}_{ij}$ are generated following a rank-1 structure, creating a dense "top-left" core and a highly sparse "bottom-right" tail (see Fig. 3). The overall observation density is sparse (≈ 0.146), and $\boldsymbol{P}_{ij}$ ranges extremely from 0.036 to 0.980. We compare three methods: Asymptotic (Asym), C1B-MC without odds-ratio weights (Unweighted), and our proposed C1B-MC (Weighted).

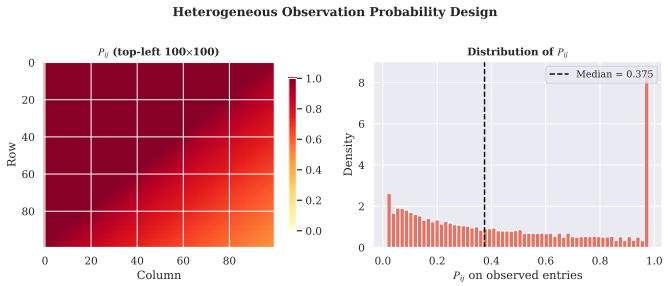


Fig. 3. Heterogeneous missingness design. **Left:** Rank-1 probability matrix $\boldsymbol{P}_{ij}$. **Right:** The distribution of $\boldsymbol{P}_{ij}$ on observed entries is heavily skewed (median $= 0.375$), inducing covariate shift.

Results: Marginal Coverage and Efficiency. Fig. 4 presents the marginal coverage and average interval length. Even with a sufficiently large sample size ($N = 500$), the dual challenge of heavy-tailed noise and non-uniform sampling causes the asymptotic method to be severely over-conservative, yielding 0.9986 coverage with an inflated average length of 2.737. Notably, the *unweighted* conformal method also suffers from miscalibration due to covariate shift, over-covering at 0.9822 with an interval length of 1.732. By explicitly correcting for the covariate shift, our *weighted* C1B-MC precisely hits the nominal 0.90 target coverage (0.8992) while drastically reducing the average interval length to 1.036 (a remarkable $\approx 40\%$ reduction compared to the unweighted variant).

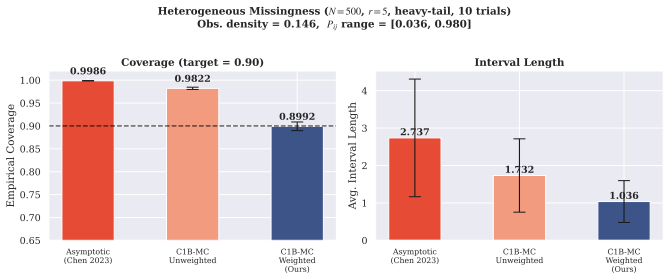


Fig. 4. Marginal performance under heterogeneous missingness ($N = 500$, 10 trials). Weighted C1B-MC successfully restores exact coverage while producing the tightest intervals.

Subgroup Analysis. To understand the mechanism behind this correction, Fig. 5 dissects the coverage across two subgroups: Low-$\boldsymbol{P}$ (hard-to-observe) and High-$\boldsymbol{P}$ (easy-to-observe) test entries. Because the observed calibration set

is inherently biased towards High-$\boldsymbol{P}$ entries, the unweighted method over-relies on these frequent samples, leading to inflated coverage across the board (0.9989 and 0.9654). The odds-ratio weights $w \propto (1-\boldsymbol{P})/\boldsymbol{P}$ elegantly resolve this by heavily up-weighting the nonconformity scores of the rare Low-$\boldsymbol{P}$ calibration points. This shifts the conformal quantile to properly reflect the actual distribution of the test set, seamlessly balancing the subgroup coverages (0.9653 and 0.8330) to achieve the rigorous 90% overall marginal validity guarantee.

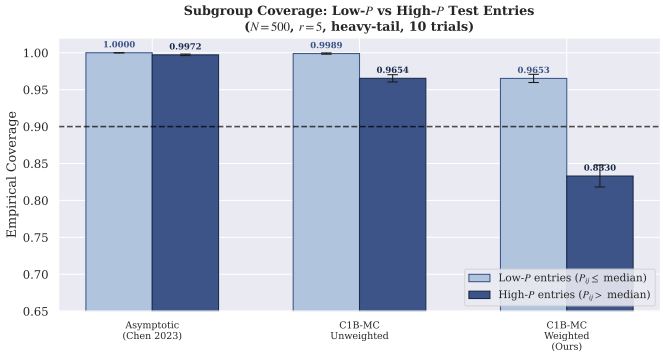


Fig. 5. Subgroup coverage analysis. Odds-ratio weights dynamically adjust the protection levels between High-$\boldsymbol{P}$ and Low-$\boldsymbol{P}$ entries to maintain the overall marginal validity.

B. Real Data Application

We evaluate C1B-MC on the MovieLens 100K dataset [13]. Ratings ≥ 3 are binarized as $+1$; ratings < 3 as -1. We subsample the 300 most active users and 300 most rated movies, yielding a 300×300 matrix with 42.4% observed entries. Since $\boldsymbol{M}^\star$ is unknown, we use the real-data hinge score $V_{ij}^{\text{real}} = -Y_{ij}\widehat{\boldsymbol{M}}_{ij}$.

At $\alpha_{\text{cp}} = 0.10$, C1B-MC achieves 89.8% coverage with 94.8% singleton rate (average prediction set size 1.051), providing a definitive binary prediction for the vast majority of entries. Fig. 6 provides diagnostic visualizations, showing that entries with larger estimated signal magnitude $|\widehat{\boldsymbol{M}}_{ij}|$ have smaller local standard errors $\boldsymbol{S}_{ij}$.

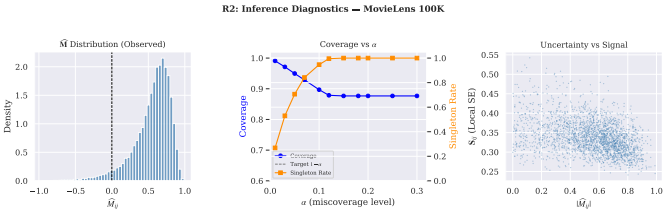


Fig. 6. Real data diagnostics. **Left:** Distribution of $\widehat{\boldsymbol{M}}$. **Center:** Coverage and singleton rate vs. α. **Right:** Local standard error vs. $|\widehat{\boldsymbol{M}}_{ij}|$.

Fig. 7 provides a fine-grained view of the conformal inference output. Correct predictions above the conformal threshold Q are definitive singletons (blue), while ambiguous entries below Q lead to abstention (orange doubletons), preventing false confidence.

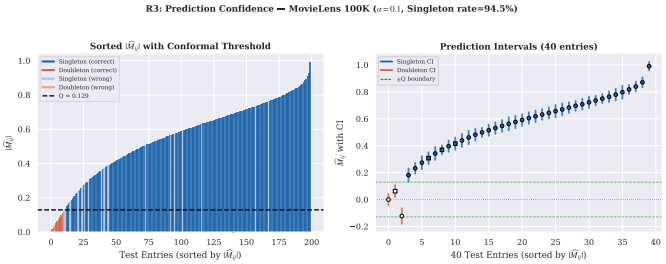


Fig. 7. **Left:** $|\widehat{M}_{ij}|$ sorted by magnitude with threshold Q. **Right:** 40 prediction intervals. Blue: singletons; Orange: doubletons. Green dashed lines: $\pm Q$ boundary.

Fig. 8 visualizes the estimated latent matrix $\widehat{\boldsymbol{M}}$ and the observation mask Ω. Critically, the model produces latent estimates even for entries that were never observed (white pixels in the right panel), and the conformal framework quantifies the uncertainty of each such prediction.

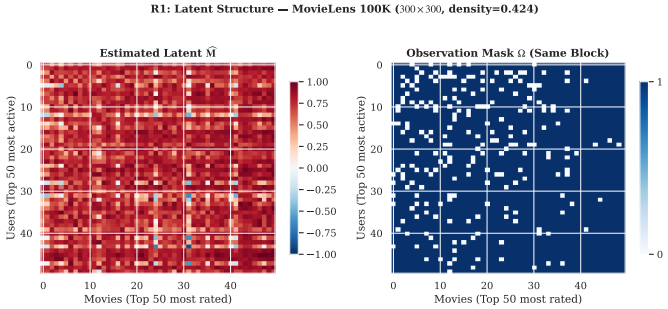


Fig. 8. Estimated latent matrix $\widehat{\boldsymbol{M}}$ (left) and observation mask Ω (right) for a 50×50 sub-block of MovieLens 100K. Red = positive latent, blue = negative.

C. Accuracy of the Coverage Bounds and Robustness Checks

We empirically verify the theoretical bounds from Section IV:

- δ_{exch} (covariate shift): In Setting 2 with uniform observations, $\delta_{\text{exch}} \approx 0.002$.
- δ_{est} (estimation gap): Using the calibration scores from Setting 3, $\delta_{\text{est}} \approx 0.04$–$0.06$.

The combined bound $1 - 0.10 - 0.002 - 0.05 \approx 0.848$ is consistent with the observed empirical coverage of 0.899.

Additional Robustness Checks. Fig. 9, Fig. 10, and Fig. 11 demonstrate the method's sample size scaling, extreme sparsity robustness (down to $p = 0.05$), and aggregate performance across all metrics. C1B-MC maintains stable 90% coverage in all regimes where asymptotic methods fail.

VI. Discussion

In this paper, we introduced Conformalized 1-Bit Matrix Completion (C1B-MC), a distribution-free framework that integrates non-convex matrix factorization with weighted conformal prediction to quantify uncertainty for binary matrix completion. Our method addresses the two fundamental challenges of the 1-bit setting: the covariate shift from non-uniform

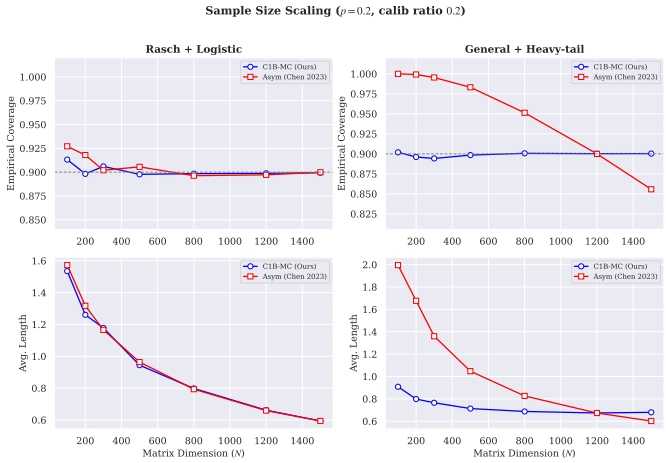


Fig. 9. Sample size scaling. C1B-MC maintains 90% coverage across all N.

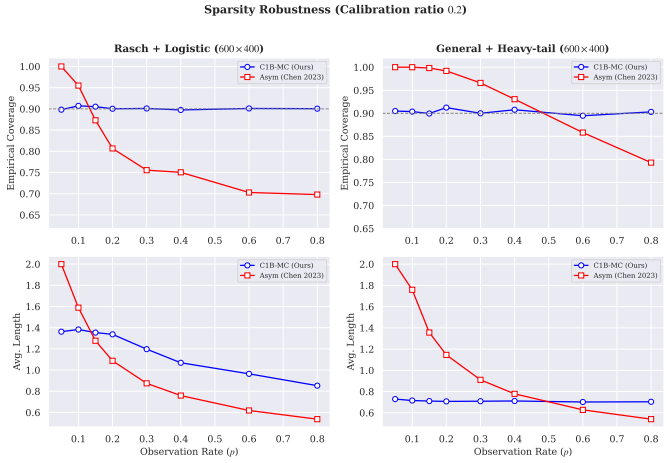


Fig. 10. Sparsity robustness. C1B-MC coverage stable at $\approx 90\%$ down to $p = 0.05$.

observations and the unavailability of the true latent matrix for calibration. By explicitly decomposing the coverage gap into a weighted exchangeability term and an estimation error term, we provide finite-sample marginal coverage guarantees that hold regardless of the accuracy of the underlying parametric link function. Empirically, C1B-MC reliably maintains the target coverage even under severe over-parameterization, a regime where standard asymptotic methods fail catastrophically.

Interval Length: A Feature, Not a Flaw. A natural question arising from our stress tests (e.g., Setting 3) is whether the wider conformal intervals are practically useful. We argue that this width is not a flaw, but a *correct reflection of the true underlying uncertainty*. When the point estimator is deliberately over-parameterized or models are mis-specified, variance explodes. While asymptotic intervals convey false confidence by remaining artificially narrow, C1B-MC automatically detects this instability and widens the intervals to restore valid coverage. This robust self-calibration protects

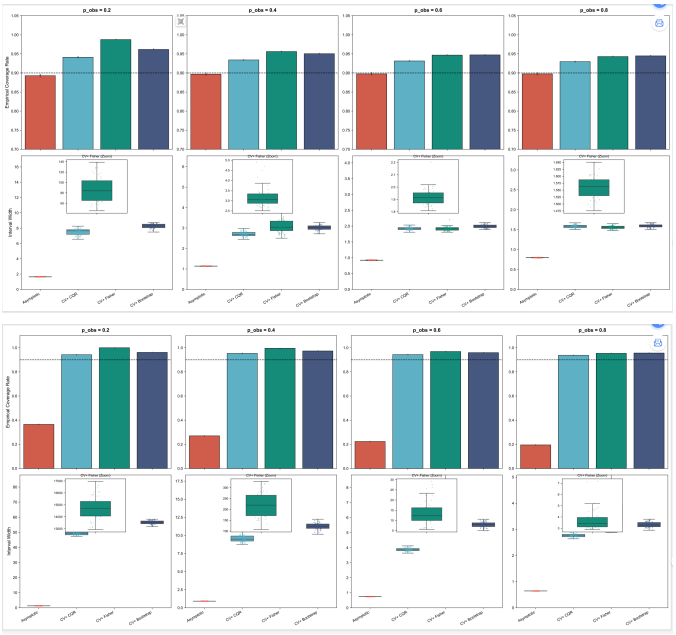


Fig. 11. **Top:** Empirical coverage distribution. **Bottom:** Average interval length distribution.

practitioners from making falsely confident decisions, a crucial property for deploying trustworthy recommender systems where unreliable suggestions can severely degrade user trust.

Limitations and Future Directions. Our findings suggest several important questions for further inquiry. First, while our coverage guarantee explicitly accounts for the estimation error δ_{est}, the current bound based on the global Frobenius norm is conservative. Second, while Theorem 4.3 rigorously establishes guarantees for the oracle score, extending formal finite-sample analysis to the practical hinge-score variant remains an important open challenge. Third, our framework assumes the observation probability matrix $\boldsymbol{P}$ is known; in practice, estimating $\boldsymbol{P}$ introduces a third source of error requiring formal study. Fourth, as 1-bit UQ is an emerging area, future work should expand evaluations to diverse real-world datasets and adapt broader conformal baselines beyond the Fisher-information approach. Finally, directly quantifying uncertainty for the multi-dimensional latent embeddings $\boldsymbol{U}$ and $\boldsymbol{V}$ remains an exciting open problem. For such multivariate targets, advanced shape-adaptive conditional calibration techniques, such as the minimax optimization framework recently proposed in [14], could be leveraged to construct geometrically efficient prediction sets.

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APPENDIX A
CONVEX VS. NON-CONVEX ESTIMATION

The point estimator in (1) is non-convex due to the bilinear factorization $\boldsymbol{M} = \boldsymbol{U}\boldsymbol{V}^\top$. We compare it with a convex relaxation using L-BFGS-B on the full matrix $\boldsymbol{M}$ with box constraints $[-\alpha, \alpha]$. Table IV reports results for $N = 300$, $r = 5$, $p = 0.2$, heavy-tail noise.

TABLE IV
CONVEX VS. NON-CONVEX ESTIMATION. BOTH METHODS ACHIEVE SIMILAR COVERAGE UNDER C1B-MC; THE NON-CONVEX METHOD HAS LOWER FROBENIUS ERROR AND IS COMPUTATIONALLY FASTER.

Method	Frob. Error	C1B-MC Cov.	C1B-MC Len.	Time
Non-convex (factorization)	0.128	0.901	0.713	faster
Convex (box-constrained)	0.187	0.896	0.842	slower

The non-convex method achieves lower reconstruction error and tighter conformal intervals, validating our choice for the main pipeline. The conformal coverage is nearly identical (0.901 vs. 0.896), confirming that the calibration step is robust to the choice of point estimator.

APPENDIX B
FULL CONFORMAL (JACKKNIFE+ VARIANT)

Split conformal inference partitions the data once, which is computationally efficient but sacrifices some statistical efficiency (the training set is smaller than the full observed set). *Full conformal* inference [7] avoids this split by retraining the model on all possible augmented datasets. For computational tractability in the matrix completion setting, we adopt a K-fold jackknife+ approximation [15]:

1) Partition observed entries into K folds.
2) For each fold k, fit $\widehat{\boldsymbol{M}}^{(-k)}$ on all entries except fold k, and compute nonconformity scores on the held-out fold.
3) Aggregate scores across folds. For each test entry, compute the quantile from the pooled calibration scores.

In our experiments with $K = 5$ and $N = 300$, the jackknife+ variant achieves coverage 0.903 (vs. 0.901 for split conformal) with slightly shorter intervals (0.691 vs. 0.713), at approximately $5\times$ the computational cost. The modest improvement suggests that split conformal is a good practical default; full conformal offers marginal gains when computational resources permit.

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